

TNB ICOMS 2.0

DATABASE DESIGN DOCUMENT

Integrated Outage Management System — Version 2.0

Technology Stack

Backend: .NET 8 (ASP.NET Core Web API) | Frontend: Blazor WebAssembly / Server
Database: Microsoft SQL Server 2022 | Auth: Active Directory (internal) + ASP.NET Membership (external)
Deployment: IIS / Azure App Service | ORM: Entity Framework Core 8

Prepared for: Tenaga Nasional Berhad (TNB) Grid System Operation

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1. Introduction

1.1 Purpose

This document provides the complete database design for the TNB ICOMS 2.0 system — a reinvention of the legacy Integrated Outage Management System. It covers entity definitions, column specifications, relationships, indexing strategy, authentication design, and data governance guidelines.

The design is aligned with the four User Requirements Specification (URS) modules: System Setup (Module 1), Requestor/Planner (Module 2), TOMS/GNM (Module 3), and GNC (Module 4).

1.2 System Overview

ICOMS 2.0 manages the full lifecycle of electrical network outages across TNB's transmission zones. Core capabilities include:

- Outage request intake, study, approval, authorisation, and closure for Planned, Unplanned, Emergency, and Forced outage types.
- Role-based and zone-based access control with Active Directory integration for internal users and ASP.NET Membership for external (GCU) users.
- Automated outage type classification based on voltage level, work type, and scheduled duration.
- Conflict detection for transmission lines, SLD management, Commissioning Memo generation, and shift Handover tracking.
- KIV management, change request workflows, off-point management, and rich statistics/reporting.

1.3 Technology Stack

Layer	Technology
Backend Framework	.NET 10.0 (backward support to 8.0) — ASP.NET Core Web API (RESTful + SignalR for real-time)
Frontend	Blazor WebAssembly / Blazor Server (with Razor Components)
Database (RDS)	Microsoft SQL Server 2022 (Azure SQL or on-premises)
ORM	Entity Framework Core 8 with Code-First Migrations
Internal Auth	Windows Authentication + Active Directory (LDAP/Kerberos via AD integration)
External Auth	ASP.NET Core Identity (Membership) — JWT Bearer tokens
API Security	OAuth 2.0 / OpenID Connect via Azure AD or on-premises ADFS
File Storage	Azure Blob Storage / Network File Share (SLDs, PDFs, attachments)
Caching	IMemoryCache / Redis for session and list caching
Logging	Serilog to SQL + Azure Application Insights

1.4 Naming Conventions

- Tables: PascalCase singular noun (e.g., Outage, Equipment, Zone).
- Primary Keys: <TableName>Id — INT IDENTITY(1,1) or UNIQUEIDENTIFIER where distributed identity is needed.
- Foreign Keys: <ReferencedTableName>Id (e.g., Zoneld, Outageld).

- Audit columns present on all write-enabled tables: CreatedAt, CreatedBy, UpdatedAt, UpdatedBy, IsDeleted (soft-delete).
- Enum-like lookups backed by dedicated tables, not hard-coded integers.
- Schema separation: dbo (core), auth (users/roles), audit (change log), config (system setup), handover (shift module).

2. Authentication & User Schema (auth)

The auth schema covers both internal users (AD-linked) and external users (GCU via ASP.NET Identity Membership). Both user types share a single Users table discriminated by an AuthType flag, enabling a unified access control model across the application.

2.1 Users

Central user record. Internal users are resolved via AD (TNB ID). External users are provisioned via ASP.NET Identity with email/password credentials stored in AspNetUsers.

auth.Users

Master user table for all system roles.

Column Name	Data Type	Key / Null	Constraint	Description
UserId	INT	PK, NOT NULL	IDENTITY(1,1)	Surrogate primary key
TnblId	NVARCHAR(20)	UNIQUE, NULL	NULL for external users	TNB Active Directory employee ID
FullName	NVARCHAR(200)	NOT NULL		User full legal name
Email	NVARCHAR(256)	NOT NULL	UNIQUE INDEX	Email address; used as login for external users
PhoneNumber	NVARCHAR(30)	NULL		Contact phone number
AuthType	TINYINT	NOT NULL	1=AD, 2=Membership	Determines authentication pathway
AspNetUserId	NVARCHAR(128)	NULL	FK → AspNetUsers.Id	Linked ASP.NET Identity record for external users
RoleId	INT	NOT NULL	FK → auth.Roles.RoleId	Assigned system role
ZonId	INT	NOT NULL	FK → config.Zones.ZonId	Assigned operational zone
OrganisationId	INT	NULL	FK → config.Organisations.OrgId	Assigned organisation
GcuTypeId	INT	NULL	FK → config.DropDownValues.Id	GCU sub-type (GCU role only)
IsActive	BIT	NOT NULL	DEFAULT 1	Account active flag
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	Record creation timestamp
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	User who created this record
UpdatedAt	DATETIME2	NULL		Last modification timestamp
UpdatedBy	INT	NULL	FK → auth.Users.UserId	User who last modified

Column Name	Data Type	Key / Null	Constraint	Description
IsDeleted	BIT	NOT NULL	DEFAULT 0	Soft-delete flag

2.2 Roles

auth.Roles

System roles seeded at installation. Configurable by SysAdmin.

Column Name	Data Type	Key / Null	Constraint	Description
RoleId	INT	PK, NOT NULL	IDENTITY(1,1)	Surrogate PK
RoleName	NVARCHAR(50)	NOT NULL	UNIQUE	Role display name (e.g., Requestor, GNC, GNM)
RoleCode	NVARCHAR(20)	NOT NULL	UNIQUE	Short code used in business logic (e.g., GCU, GNM_ADMIN)
IsExternal	BIT	NOT NULL	DEFAULT 0	TRUE for external-facing roles (GCU)
IsActive	BIT	NOT NULL	DEFAULT 1	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

2.3 RolePermissions

auth.RolePermissions

Granular feature permissions per role, configurable from Admin UI.

Column Name	Data Type	Key / Null	Constraint	Description
PermissionId	INT	PK, NOT NULL	IDENTITY(1,1)	
RoleId	INT	NOT NULL	FK → auth.Roles.RoleId	
FeatureCode	NVARCHAR(100)	NOT NULL		Feature identifier (e.g., OUTAGE_CREATE, OUTAGE_APPROVE)
CanView	BIT	NOT NULL	DEFAULT 0	
CanCreate	BIT	NOT NULL	DEFAULT 0	
CanEdit	BIT	NOT NULL	DEFAULT 0	
CanDelete	BIT	NOT NULL	DEFAULT 0	
CanApprove	BIT	NOT NULL	DEFAULT 0	

Column Name	Data Type	Key / Null	Constraint	Description
UpdatedAt	DATETIME2	NULL		
UpdatedBy	INT	NULL	FK → auth.Users.UserId	

2.4 UserGcuStations

auth.UserGcuStations

Many-to-many mapping of GCU users to stations they are authorised to create outages for.

Column Name	Data Type	Key / Null	Constraint	Description
UserGcuStationId	INT	PK, NOT NULL	IDENTITY(1,1)	
UserId	INT	NOT NULL	FK → auth.Users.UserId	
StationId	INT	NOT NULL	FK → config.Stations.StationId	

2.5 RoleTransferRequests

auth.RoleTransferRequests

User-initiated requests to change role or zone assignment, reviewed by SysAdmin.

Column Name	Data Type	Key / Null	Constraint	Description
TransferRequestId	INT	PK, NOT NULL	IDENTITY(1,1)	
UserId	INT	NOT NULL	FK → auth.Users.UserId	Requesting user
RequestedRoleId	INT	NULL	FK → auth.Roles.RoleId	New role requested
RequestedZoneId	INT	NULL	FK → config.Zones.ZoneId	New zone requested
RequestSummary	NVARCHAR(MAX)	NULL		Free-text reason for transfer
StatusCode	NVARCHAR(20)	NOT NULL	PENDING/APPROVED/REJECTED	
ReviewedBy	INT	NULL	FK → auth.Users.UserId	SysAdmin reviewer
ReviewedAt	DATETIME2	NULL		
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

3. System Configuration Schema (config)

The config schema stores all master data maintained by GNM ADMIN and other administrative roles. These tables drive dropdowns, hierarchies, and validation rules throughout the application.

3.1 Zones

config.Zones

Geographical operational zones (e.g., KL, Eastern, Northern).

Column Name	Data Type	Key / Null	Constraint	Description
Zoneld	INT	PK, NOT NULL	IDENTITY(1,1)	
ZoneName	NVARCHAR(100)	NOT NULL	UNIQUE	e.g., KL, Eastern, Northern
ZoneAbbr	NVARCHAR(10)	NOT NULL	UNIQUE	Short code
IsActive	BIT	NOT NULL	DEFAULT 1	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	
UpdatedAt	DATETIME2	NULL		
UpdatedBy	INT	NULL	FK → auth.Users.UserId	

3.2 ZoneLocations

config.ZoneLocations

Maps Malaysian states/territories to zones.

Column Name	Data Type	Key / Null	Constraint	Description
ZoneLocationId	INT	PK, NOT NULL	IDENTITY(1,1)	
Zoneld	INT	NOT NULL	FK → config.Zones.Zoneld	
LocationName	NVARCHAR(100)	NOT NULL		e.g., Selangor, Kelantan
IsActive	BIT	NOT NULL	DEFAULT 1	

3.3 Organisations

config.Organisations

Internal and external organisations mapped to zones.

Column Name	Data Type	Key / Null	Constraint	Description
OrgId	INT	PK, NOT NULL	IDENTITY(1,1)	
OrgName	NVARCHAR(200)	NOT NULL		Organisation display name
OrgAbbr	NVARCHAR(20)	NOT NULL	UNIQUE	Abbreviation — must be globally unique
Zoneld	INT	NOT NULL	FK → config.Zones.Zoneld	
IsExternal	BIT	NOT NULL	DEFAULT 0	TRUE for external (e.g., GCU) organisations
IsActive	BIT	NOT NULL	DEFAULT 1	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	

3.4 Stations

config.Stations

Electrical stations within zones and organisations.

Column Name	Data Type	Key / Null	Constraint	Description
StationId	INT	PK, NOT NULL	IDENTITY(1,1)	
StationName	NVARCHAR(200)	NOT NULL		
StationAbbr	NVARCHAR(20)	NOT NULL	UNIQUE	Must be globally unique
Zoneld	INT	NOT NULL	FK → config.Zones.Zoneld	
OrgId	INT	NOT NULL	FK → config.Organisations.OrgId	
SldFileUrl	NVARCHAR(500)	NULL	foreign key reference since one station can have multiple SLDs	URL to current Single Line Diagram PDF
IsActive	BIT	NOT NULL	DEFAULT 1	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	
UpdatedAt	DATETIME2	NULL		

Column Name	Data Type	Key / Null	Constraint	Description
UpdatedBy	INT	NULL	FK → auth.Users.UserId	

3.5 VoltageLevel

config.VoltageLevel

System-wide voltage levels (e.g., 132kV, 275kV, 500kV).

Column Name	Data Type	Key / Null	Constraint	Description
VoltageLevelId	INT	PK, NOT NULL	IDENTITY(1,1)	
VoltageName	NVARCHAR(50)	NOT NULL	UNIQUE	e.g., 132kV, 275kV, 500kV
VoltageKv	DECIMAL(8,2)	NOT NULL		Numeric kV value for rule matching
DisplayOrder	INT	NOT NULL	DEFAULT 0	Sort order in dropdowns
IsActive	BIT	NOT NULL	DEFAULT 1	

3.6 EquipmentType

config.EquipmentType

Equipment types tied to voltage levels (e.g., Busbar, Line, Transformer).

Column Name	Data Type	Key / Null	Constraint	Description
EquipmentTypeId	INT	PK, NOT NULL	IDENTITY(1,1)	
VoltageLevelId	INT	NOT NULL	FK → config.VoltageLevel.VoltageLevelId	
TypeName	NVARCHAR(100)	NOT NULL		e.g., Line, SGT, Busbar, Bay
TypeAbbr	NVARCHAR(20)	NULL		
IsActive	BIT	NOT NULL	DEFAULT 1	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	

3.7 Equipment

config.Equipment

Individual equipment items. Equipment name auto-generated as 'VoltageLevel – MVA – Name'.

Column Name	Data Type	Key / Null	Constraint	Description
EquipmentId	INT	PK, NOT NULL	IDENTITY(1,1)	
EquipmentName	NVARCHAR(300)	NOT NULL	Auto-generated: VL–MVA–Name	Display name
EquipmentCode	NVARCHAR(50)	NOT NULL	UNIQUE	System-generated code
EquipmentTypeId	INT	NOT NULL	FK → config.EquipmentType.EquipmentTypeId	
VoltageLevelId	INT	NOT NULL	FK → config.VoltageLevel.VoltageLevelId	
StationId	INT	NOT NULL	FK → config.Stations.StationId	
ZoneId	INT	NOT NULL	FK → config.Zones.ZoneId	Denormalised for query perf
MvaRatingId	INT	NULL	FK → config.DropdownValues.Id	Selected MVA rating
Position	TINYINT	NOT NULL	0=Closed, 1=Open	Current switch position
IsOffPoint	BIT	NOT NULL	DEFAULT 0	Permanent off-point flag
OffPointRemark	NVARCHAR(MAX)	NULL		Reason for off-point
LineFilterType	NVARCHAR(20)	NULL	Single/Tee-Off/Quad/NULL	For line equipment only
IsActive	BIT	NOT NULL	DEFAULT 1	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	
UpdatedAt	DATETIME2	NULL		
UpdatedBy	INT	NULL	FK → auth.Users.UserId	

3.8 ConflictingLines

config.ConflictingLines

Pairs of equipment (transmission lines) that logically conflict during outage scheduling.

Column Name	Data Type	Key / Null	Constraint	Description
ConflictId	INT	PK, NOT NULL	IDENTITY(1,1)	
PrimaryEquipmentId	INT	NOT NULL	FK → config.Equipment.EquipmentId	
ConflictEquipmentId	INT	NOT NULL	FK → config.Equipment.EquipmentId	
ConflictReason	NVARCHAR(500)	NULL		Optional explanation
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	

3.9 DropdownValues

config.DropdownValues

Generic configurable dropdown store. Admin-managed. Covers JobType, WorkType, Sequence, Restoration, GcuType, MvaRating, NotTakenReason, etc.

Column Name	Data Type	Key / Null	Constraint	Description
DropdownValueId	INT	PK, NOT NULL	IDENTITY(1,1)	
Category	NVARCHAR(50)	NOT NULL		e.g., JobType, WorkType, Restoration, GcuType
ValueCode	NVARCHAR(50)	NOT NULL		Internal code
ValueText	NVARCHAR(200)	NOT NULL		Display label
ParentCode	NVARCHAR(50)	NULL		For JobType: OutageTypeCode (Planned/Unplanned/Emergency/Forced)
DisplayOrder	INT	NOT NULL	DEFAULT 0	
IsActive	BIT	NOT NULL	DEFAULT 1	
UpdatedAt	DATETIME2	NULL		

Column Name	Data Type	Key / Null	Constraint	Description
UpdatedBy	INT	NULL	FK → auth.Users.UserId	

3.10 Projects

config.Projects

Project records created by Project Engineers. Linked to outages (non-maintenance job types only).

Column Name	Data Type	Key / Null	Constraint	Description
ProjectId	INT	PK, NOT NULL	IDENTITY(1,1)	
TpCode	NVARCHAR(50)	NOT NULL	UNIQUE	Level-4 TP code — unique, non-reusable
ProjectSuffix	NVARCHAR(200)	NOT NULL		Reusable project name suffix
ProjectName	NVARCHAR(260)	NOT NULL	Computed: TpCode + ' – ' + ProjectSuffix	Display name
Status	TINYINT	NOT NULL	0=Open, 1=Closed	
StartDate	DATE	NULL	Auto-filled from first linked outage	
EndDate	DATE	NULL	Auto-filled on project closure	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	Project Engineer
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
ClosedBy	INT	NULL	FK → auth.Users.UserId	GNM ADMIN who closed
ClosedAt	DATETIME2	NULL		

3.11 OutageTypeRules

config.OutageTypeRules

Configurable rules that auto-assign outage type (Planned/Unplanned/Emergency/Forced) based on work type, voltage, and lead time.

Column Name	Data Type	Key / Null	Constraint	Description
RuleId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageTypeCode	NVARCHAR(20)	NOT	Planned/Unplanned/Emergency/Forced	

Column Name	Data Type	Key / Null	Constraint	Description
		NULL		
WorkTypeCode	NVARCHAR(20)	NOT NULL	Dead/Live	
MoreThanDays	INT	NULL		Lead time lower bound (days)
MoreThanMonths	INT	NULL		Lead time lower bound (months)
MoreThanYears	INT	NULL		Lead time lower bound (years)
LessThanDays	INT	NULL		Lead time upper bound (days)
LessThanMonths	INT	NULL		Lead time upper bound (months)
LessThanYears	INT	NULL		Lead time upper bound (years)
AppliesTo	NVARCHAR(50)	NOT NULL	ALL / comma-separated kV values	Voltage filter
IsActive	BIT	NOT NULL	DEFAULT 1	
UpdatedAt	DATETIME2	NULL		
UpdatedBy	INT	NULL	FK → auth.Users.UserId	

3.12 AuthorisationPersonnel

config.AuthorisationPersonnel

Pre-approved signatories for outage authorisations, maintained by GNM ADMIN.

Column Name	Data Type	Key / Null	Constraint	Description
PersonnelId	INT	PK, NOT NULL	IDENTITY(1,1)	
FullName	NVARCHAR(200)	NOT NULL		
Email	NVARCHAR(256)	NOT NULL		Notified on Taken-Active / Taken-Completed status
Zoneld	INT	NOT NULL	FK → config.Zones.Zoneld	
PositionTitle	NVARCHAR(100)	NULL		
IsActive	BIT	NOT NULL	DEFAULT 1	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

4. Outage Schema (dbo)

The outage schema forms the operational core of ICOMS 2.0. It tracks every outage from creation through to closure, along with all workflow statuses, study notes, change requests, and PIC records.

4.1 Outages

dbo.Outages

Master outage record. One row per outage request, regardless of type or lifecycle stage.

Column Name	Data Type	Key / Null	Constraint	Description
OutageId	INT	PK, NOT NULL	IDENTITY(1,1)	Surrogate PK
OutageNumber	NVARCHAR(30)	NOT NULL	UNIQUE — auto-generated	Human-readable ID (e.g., ICOMS-2024-00001)
OutageCode	CHAR(1)	NOT NULL	P/U/E/F — auto-set	Planned/Unplanned/Emergency/Forced
OutageTypeCode	NVARCHAR(20)	NOT NULL	Planned/Unplanned/Emergency/Forced	Auto-classified via OutageTypeRules
OutageClass	NVARCHAR(20)	NOT NULL	Maintenance/Project	Maintenance if JobType=Maintenance, else Project
WorkTypeCode	NVARCHAR(10)	NOT NULL	Dead/Live — FK DropdownValues	
Zoneld	INT	NOT NULL	FK → config.Zones.Zoneld	
StationId	INT	NOT NULL	FK → config.Stations.StationId	
VoltageLevelId	INT	NOT NULL	FK → config.VoltageLevel.VoltageLevelId	
PrimaryEquipmentId	INT	NOT NULL	FK → config.Equipment.EquipmentId	Main equipment
LineFilterType	NVARCHAR(20)	NULL	Single/Tee-Off/Quad	
JobTypeId	INT	NOT NULL	FK → config.DropdownValues.DropdownValueId	

Column Name	Data Type	Key / Null	Constraint	Description
ProjectId	INT	NUL L	FK → config.Projects.ProjectId	Only for non-Maintenance outages
SequenceId	INT	NUL L	FK → config.DropDownValues.DropDownValued	
RestorationId	INT	NUL L	FK → config.DropDownValues.DropDownValued	
PlannedStartAt	DATETIME2	NOT NUL L		Scheduled outage start
PlannedEndAt	DATETIME2	NOT NUL L		Scheduled outage end
ActualStartAt	DATETIME2	NUL L		Set when Taken-Active
ActualEndAt	DATETIME2	NUL L		Set when Outage Closed
ExtendedEndAt	DATETIME2	NUL L		GNC extension timestamp
HasPtw	BIT	NOT NUL L	DEFAULT 0	Permit to Work flag — notifies GNM daily
Description	NVARCHAR(MAX)	NUL L		Free-text outage description
ContingencyPlanUrl	NVARCHAR(500)	NUL L		DSO-uploaded PDF URL
RequestorStatus	NVARCHAR(30)	NOT NUL L	See status matrix	Requestor-side status
PlannerStatus	NVARCHAR(30)	NUL L	See status matrix	Planner-side status (NULL for Emergency)
GnmStatus	NVARCHAR(30)	NUL L	See status matrix	GNM/TOMS status
GncStatus	NVARCHAR(30)	NUL L	See status matrix	GNC execution status
DsoAgreedAt	DATETIME2	NUL L		DSO agreement timestamp (auto-agreed after 7 days if NULL)
DsoAgreedBySystem	BIT	NOT NUL L	DEFAULT 0	Flagged when system auto-agrees on behalf of DSO
Justification	NVARCHAR(MAX)	NUL L		TOMS study field
Highlights	NVARCHAR(MAX)	NUL L		TOMS study field

Column Name	Data Type	Key / Null	Constraint	Description
Remark	NVARCHAR(MAX)	NULL		TOMS study field
UnderStudyNotes	NVARCHAR(MAX)	NULL		TOMS study field
NotTakenReasonId	INT	NULL	FK → config.DropDownValues.DropDownValueId	Required if Outage Closed - Not Taken
ApprovedById	INT	NULL	FK → config.AuthorisationPersonnel.PersonnelId	GNC forced outage approver
IsGcuNotified	BIT	NOT NULL	DEFAULT 0	GNM pushed notification to GCU
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
UpdatedBy	INT	NULL	FK → auth.Users.UserId	
UpdatedAt	DATETIME2	NULL		
IsDeleted	BIT	NOT NULL	DEFAULT 0	

4.2 Outage Status Reference

The following status values are used across the four status columns on the Outages table. Business logic in the .NET service layer enforces valid transitions.

Column	Status Value	Meaning / Trigger
RequestorStatus	Draft	Created but not yet submitted
RequestorStatus	Pending	Submitted by Requestor; awaiting Planner agreement
RequestorStatus	Confirmed	Requestor confirmed after Planner agreed
RequestorStatus	Outage Closed – Unconfirmed	Auto-set if not confirmed 2 weeks before start
PlannerStatus	Agreed	Planner (or DSO or system auto-agree) agreed
PlannerStatus	Disagreed	Planner rejected; reverts Requestor to Draft
GnmStatus	Pending	Awaiting TOMS review
GnmStatus	Under-Study	TOMS actively studying the outage
GnmStatus	KIV	Kept In View — removed from active pool

Column	Status Value	Meaning / Trigger
GnmStatus	Approved	GNM approved; available for GNC execution
GnmStatus	Disapproved	GNM rejected; outage ends
GnmStatus	Outage Closed – KIV Timeout	KIV exceeded 1 month; auto-closed
GncStatus	Taken-Active	GNC authorised and commenced outage
GncStatus	Taken-Completed	Authorisation closed; outage complete segment
GncStatus	Outage Closed	GNC formally closed the outage
GncStatus	Outage Closed – Not Taken	Approved outage not actioned within 5 days of end date
GncStatus	Outage Closed – Cancelled by GNC	GNC cancelled; Requestor may re-submit same ID

4.3 OutageAdditionalEquipment

dbo.OutageAdditionalEquipment

Supports multi-select additional equipment per outage.

Column Name	Data Type	Key / Null	Constraint	Description
OutageEquipmentId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageId	INT	NOT NULL	FK → dbo.Outages.OutageId	
EquipmentId	INT	NOT NULL	FK → config.Equipment.EquipmentId	
EquipmentTypeId	INT	NOT NULL	FK → config.EquipmentType.EquipmentTypeId	
DisplayOrder	INT	NOT NULL	DEFAULT 0	

4.4 OutagePic

dbo.OutagePic

Person-In-Charge contacts for an outage. All PICs are emailed on outage creation and on 'Update and Notify'.

Column Name	Data Type	Key / Null	Constraint	Description
PicId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageId	INT	NOT NULL	FK → dbo.Outages.OutageId	
FullName	NVARCHAR(200)	NOT NULL		
Email	NVARCHAR(256)	NOT NULL		

Column Name	Data Type	Key / Null	Constraint	Description
PhoneNumber	NVARCHAR(30)	NULL		
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

4.5 OutageNotifyEmails

dbo.OutageNotifyEmails

Persistent additional email recipients for 'Update and Notify' — saved per outage so they default next time.

Column Name	Data Type	Key / Null	Constraint	Description
NotifyEmailId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageId	INT	NOT NULL	FK → dbo.Outages.OutageId	
Email	NVARCHAR(256)	NOT NULL		

4.6 ChangeRequests

dbo.ChangeRequests

Requestor-submitted modification requests for outages in Pending / KIV status.

Column Name	Data Type	Key / Null	Constraint	Description
ChangeRequestId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageId	INT	NOT NULL	FK → dbo.Outages.OutageId	
RequestedBy	INT	NOT NULL	FK → auth.Users.UserId	
ChangeType	NVARCHAR(30)	NOT NULL	DateTimeChange/EquipmentChange/Cancellation	
NewStartAt	DATETIME2	NULL		
NewEndAt	DATETIME2	NULL		
NewEquipmentId	INT	NULL	FK → config.Equipment.EquipmentId	
ChangeReason	NVARCHAR(MAX)	NULL		
Status	NVARCHAR(20)	NOT NULL	Pending/Approved/Rejected	
ReviewedBy	INT	NULL	FK → auth.Users.UserId	

Column Name	Data Type	Key / Null	Constraint	Description
ReviewedAt	DATETIME2	NULL		
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

4.7 OutageOffPoints

dbo.OutageOffPoints

Off-point suggestions made by TOMS during outage study. Also tracks temporary off-points during execution.

Column Name	Data Type	Key / Null	Constraint	Description
OutageOffPointId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageId	INT	NOT NULL	FK → dbo.Outages.OutageId	
EquipmentId	INT	NULL	FK → config.Equipment.EquipmentId	NULL = free-text temporary
IsTemporary	BIT	NOT NULL	DEFAULT 0	
SuggestedAction	NVARCHAR(20)	NULL	Open/Close	
Remarks	NVARCHAR(MAX)	NULL		GNC-facing note
SuggestedBy	INT	NOT NULL	FK → auth.Users.UserId	TOMS user
RemarksBy	INT	NOT NULL	FK → auth.Users.UserId	GNC user
RemarksAT	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
SuggestedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
IsNormalised	BIT	NOT NULL	DEFAULT 0	Set TRUE when GNC normalises on closure

4.8 GcuAcknowledgements

dbo.GcuAcknowledgements

GCU user responses to outage notifications from GNM.

Column Name	Data Type	Key / Null	Constraint	Description
AcknowledgementId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageId	INT	NOT NULL	FK → dbo.Outages.OutageId	
GcuUserId	INT	NOT NULL	FK → auth.Users.UserId	
Response	NVARCHAR(20)	NOT NULL	Agreed/Disagreed/Informed	Informed = no

Column Name	Data Type	Key / Null	Constraint	Description
				reply received
Comments	NVARCHAR(MAX)	NULL		
AcknowledgedAt	DATETIME2	NULL		NULL if no response
NotifiedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	When GNM pushed notification

4.9 Authorisations

dbo.Authorisations

GNC outage authorisation records. Multiple per outage. Each authorisation maps to one Taken-Active / Taken-Completed cycle.

Column Name	Data Type	Key / Null	Constraint	Description
AuthorisationId	INT	PK, NOT NULL	IDENTITY(1,1)	
OutageId	INT	NOT NULL	FK → dbo.Outages.OutageId	
AuthorisationNo	NVARCHAR(50)	NOT NULL		Physical site authorisation form number
PersonnelId	INT	NOT NULL	FK → config.AuthorisationPersonnel.PersonnelId	Selected authorising person
TakenActiveAt	DATETIME2	NOT NULL		When Taken-Active status set
TakenCompletedAt	DATETIME2	NULL		When authorisation closed
Remark	NVARCHAR(MAX)	NULL		
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	GNC user
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

5. SLD Management & Commissioning Schema (dbo)

5.1 SingleLineDiagrams

dbo.SingleLineDiagrams

Versioned SLD records per station. Approval workflow managed by GNM.

Column Name	Data Type	Key / Null	Constraint	Description
SldId	INT	PK, NOT NULL	IDENTITY(1,1)	
StationId	INT	NOT NULL	FK → config.Stations.StationId	
Revision	NVARCHAR(20)	NOT NULL		e.g., Rev 1.0, Rev 2.0
FileUrl	NVARCHAR(500)	NOT NULL		Blob/file storage URL
StatusCode	NVARCHAR(20)	NOT NULL	Draft/Submitted/Approved/Published	
SubmittedBy	INT	NULL	FK → auth.Users.UserId	
SubmittedAt	DATETIME2	NULL		
ApprovedBy	INT	NULL	FK → auth.Users.UserId	
ApprovedAt	DATETIME2	NULL		
PublishedAt	DATETIME2	NULL		When made system-wide visible
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	
IsDeleted	BIT	NOT NULL	DEFAULT 0	

5.2 CommissioningMemos

dbo.CommissioningMemos

Commissioning memos linked to outages, created and approved within GNM.

Column Name	Data Type	Key / Null	Constraint	Description
Memoid	INT	PK, NOT NULL	IDENTITY(1,1)	
Outageld	INT	NOT NULL	FK → dbo.Outages.Outageld	
MemoNumber	NVARCHAR(50)	NOT NULL	UNIQUE	Auto-generated memo reference
Content	NVARCHAR(MAX)	NULL		Memo body
FileUrl	NVARCHAR(500)	NULL		Generated PDF URL
StatusCode	NVARCHAR(20)	NOT NULL	Draft/Submitted/Approved	
CreatedBy	INT	NOT NULL	FK → auth.Users.UserId	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
ApprovedBy	INT	NULL	FK → auth.Users.UserId	
ApprovedAt	DATETIME2	NULL		

6. Handover Schema (handover)

The Handover module is exclusive to GNC users. It captures shift-based operational records passed from one shift to the next across three daily shifts (Morning, Evening, Night).

6.1 HandoverShifts

handover.HandoverShifts

Master record per shift per day.

Column Name	Data Type	Key / Null	Constraint	Description
ShiftId	INT	PK, NOT NULL	IDENTITY(1,1)	
ShiftDate	DATE	NOT NULL		
ShiftType	NVARCHAR(10)	NOT NULL	Morning/Evening/Night	
Zoneld	INT	NOT NULL	FK → config.Zones.Zoneld	
ControlManager	INT	NULL	FK → auth.Users.UserId	
SwitchEngineer1	INT	NULL	FK → auth.Users.UserId	
SwitchEngineer2	INT	NULL	FK → auth.Users.UserId	
Despatcher	INT	NULL	FK → auth.Users.UserId	
ControlAssistant	INT	NULL	FK → auth.Users.UserId	
IsPassed	BIT	NOT NULL	DEFAULT 0	Locked once handed over
PassedAt	DATETIME2	NULL		
PassedBy	INT	NULL	FK → auth.Users.UserId	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

6.2 HandoverEntries

handover.HandoverEntries

Individual line items for each tab section of the Handover report.

Column Name	Data Type	Key / Null	Constraint	Description
EntryId	INT	PK, NOT NULL	IDENTITY(1,1)	
ShiftId	INT	NOT NULL	FK → handover.HandoverShifts.ShiftId	
TabCode	NVARCHAR(50)	NOT NULL	Outages/Alarms/Emergency/SystemMonitoring/etc	
FieldCode	NVARCHAR(50)	NOT NULL		Identifies the specific field within the tab
EntryText	NVARCHAR(MAX)	NULL		Free-text content
RefOutageId	INT	NULL	FK → dbo.Outages.OutageId	For outage-linked entries
DisplayOrder	INT	NOT NULL	DEFAULT 0	
EnteredBy	INT	NOT NULL	FK → auth.Users.UserId	
EnteredAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	

7. Audit & Reporting Schema (audit)

7.1 AuditLog

audit.AuditLog

Immutable append-only log of all create, update, and delete operations across write-enabled tables.

Column Name	Data Type	Key / Null	Constraint	Description
AuditId	BIGINT	PK, NOT NULL	IDENTITY(1,1)	
TableName	NVARCHAR(100)	NOT NULL		e.g., dbo.Outages
RecordId	NVARCHAR(50)	NOT NULL		Primary key value of affected record
Action	NVARCHAR(10)	NOT NULL	INSERT/UPDATE/DELETE	
ColumnName	NVARCHAR(100)	NULL		NULL for INSERT/DELETE
OldValue	NVARCHAR(MAX)	NULL		Previous value
NewValue	NVARCHAR(MAX)	NULL		New value
ChangedBy	INT	NOT NULL	FK → auth.Users.UserId	
ChangedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
IpAddress	NVARCHAR(50)	NULL		Client IP captured by API middleware

7.2 SavedReportFilters

audit.SavedReportFilters

User-saved filter presets for Outage Repository and Statistics pages.

Column Name	Data Type	Key / Null	Constraint	Description
FilterId	INT	PK, NOT NULL	IDENTITY(1,1)	
UserId	INT	NOT NULL	FK → auth.Users.UserId	
FilterName	NVARCHAR(100)	NOT NULL		User-defined label
FilterJson	NVARCHAR(MAX)	NOT NULL		Serialized filter criteria as JSON
ReportContext	NVARCHAR(50)	NOT NULL	OutageRepo/Statistics/Report	Page this preset applies to
IsDefault	BIT	NOT NULL	DEFAULT 0	
CreatedAt	DATETIME2	NOT NULL	DEFAULT GETUTCDATE()	
UpdatedAt	DATETIME2	NULL		

8. Indexing Strategy

All primary keys are clustered. The following non-clustered indexes are recommended to support the primary query patterns of ICOMS 2.0.

Table	Index Columns	Purpose
dbo.Outages	Zoneld, GnmStatus, PlannedStartAt	Zone docket queries (TOMS/GNC approval screens)
dbo.Outages	CreatedBy, OutageTypeCode, RequestorStatus	Requestor/Planner personal listings
dbo.Outages	PrimaryEquipmentId, PlannedStartAt	Equipment history and conflict detection
dbo.Outages	GncStatus, ActualStartAt	Active and scheduled outage monitoring
dbo.Outages	PlannedStartAt, PlannedEndAt (INCLUDE Zoneld)	Calendar and docket date-range filters
config.Equipment	StationId, VoltageLevelId, EquipmentTypeId	Equipment dropdowns in outage creation
config.Equipment	IsOffPoint, Zoneld	Off-point listing screens
auth.Users	TnbId	AD lookup on login
auth.Users	Email	Membership authentication
dbo.Authorisations	OutageId, TakenActiveAt	Authorisation in Force listing
dbo.GcuAcknowledgements	OutageId, GcuUserId	GCU acknowledgement checks
audit.AuditLog	TableName, RecordId, ChangedAt	Audit trail retrieval by record
handover.HandoverShifts	ShiftDate, ShiftType, Zoneld	Shift lookup on Handover page

Partitioning: Consider partitioning **dbo.Outages** and **audit.AuditLog** by year on **PlannedStartAt** / **ChangedAt** once data volume exceeds 5 million rows. Use SQL Server sliding-window partition maintenance.

9. Key Entity Relationships

9.1 Foreign Key Summary

Parent Table	FK Column	Child Table	Description
config.Zones	ZoneId	auth.Users	User zone assignment
auth.Roles	RoleId	auth.Users	User role assignment
config.Organisations	OrgId	auth.Users	User organisation
config.Zones	ZoneId	config.Organisations	Organisation zone grouping
config.Organisations	OrgId	config.Stations	Station organisational ownership
config.Zones	ZoneId	config.Stations	Station zone
config.VoltageLevel	VoltageLevelId	config.EquipmentType	Equipment types by voltage
config.EquipmentType	EquipmentTypeId	config.Equipment	Equipment classification
config.Stations	StationId	config.Equipment	Equipment location
config.Zones	ZoneId	config.Equipment	Denormalised zone for perf
dbo.Outages	OutageId	dbo.OutageAdditionalEquipment	Additional equipment list
dbo.Outages	OutageId	dbo.OutagePic	Person-in-charge records
dbo.Outages	OutageId	dbo.ChangeRequests	Modification requests
dbo.Outages	OutageId	dbo.Authorisations	GNC authorisation records
dbo.Outages	OutageId	dbo.OutageOffPoints	TOMS off-point suggestions
dbo.Outages	OutageId	dbo.GcuAcknowledgements	GCU acknowledgement records
dbo.Outages	OutageId	dbo.CommissioningMemos	Commissioning memo link
config.Stations	StationId	dbo.SingleLineDiagrams	SLD per station
handover.HandoverShifts	ShiftId	handover.HandoverEntries	Shift detail line items

9.2 Aggregate Relationship Summary

The following hierarchy illustrates the primary parent-child ownership chain for the core operational data:

- Zone → Organisations → Stations → Equipment (VoltageLevel/EquipmentType as classifiers).
- Outages → OutageAdditionalEquipment, OutagePic, OutageNotifyEmails, ChangeRequests, Authorisations, OutageOffPoints, GcuAcknowledgements, CommissioningMemos.
- Users (Role, Zone, Organisation) → Outages (CreatedBy), Authorisations, HandoverShifts.
- HandoverShifts → HandoverEntries (with optional OutageId reference).
- SingleLineDiagrams → Stations (versioned, one active per station at a time).

10. Security & Authentication Design

10.1 Internal Users — Active Directory

- Internal users (Requestor, Planner, GNM, GNC, GNM ADMIN, SysAdmin) authenticate via Windows Authentication (Kerberos/NTLM) or OpenID Connect against TNB's Active Directory / ADFS.
- On first login the API resolves the user by TnbId (samAccountName or employee number). If no matching record in auth.Users exists, a 403 is returned until SysAdmin provisions the account.
- JWT Bearer tokens are issued by the .NET API upon successful AD token validation; token lifetime is 8 hours with sliding refresh.
- Role claims are sourced from auth.RolePermissions (not AD groups) to ensure ICOMS-specific permission granularity.

10.2 External Users — ASP.NET Identity Membership

- External users (GCU — Grid Connected Users) register via email invitation sent by SysAdmin. Credentials are stored in AspNetUsers (ASP.NET Core Identity tables in the auth schema).
- Password hashing: ASP.NET Core Identity default (PBKDF2 with HMACSHA512, 100,000 iterations).
- Email confirmation is mandatory before first login.
- External users are restricted to their assigned GcuStations and can only view/acknowledge outages in those stations.
- JWT tokens for external users carry an IsExternal=true claim, which is enforced in every API controller via policy-based authorization.

10.3 Row-Level Security (Zone Isolation)

All data queries are filtered by the Zoneld claim embedded in the JWT. The .NET service layer applies a mandatory WHERE Zoneld = @UserZoneld clause on every outage, equipment, and configuration query. GNM ADMIN and SysAdmin roles are exempt and can query across all zones.

10.4 SQL Server Security

- The application connects using a dedicated low-privilege SQL login (icoms_app) with permissions scoped to the ICOMS database only.
- Stored procedures and EF Core parameterised queries are used exclusively — no dynamic SQL construction from user input.
- Transparent Data Encryption (TDE) enabled at the database level.
- Row-level encryption applied to sensitive columns (Email, PhoneNumber) using SQL Server Always Encrypted where required by TNB data classification policy.
- Audit log table (audit.AuditLog) is restricted to INSERT only for the application login; SELECT is permitted for SysAdmin only.

11. Automated Workflow Rules (Database-Backed)

The following time-based rules are enforced by .NET background services (IHostedService / Hangfire). Each rule reads/writes to the database and produces entries in audit.AuditLog.

Rule	Trigger	Action
Unconfirmed Auto-Close	2 weeks before start	Set RequestorStatus = 'Outage Closed – Unconfirmed'
DSO Auto-Agree	7 days after pushed	Set PlannerStatus = 'Agreed', DsoAgreedBySystem = 1
KIV Auto-Timeout	31 days in KIV	Set GnmStatus = 'Outage Closed – KIV Timeout'
Not Taken Auto-Close	5 days after end date	Set GncStatus = 'Outage Closed – Not Taken' if no Taken-Active
Pending Auto-KIV	24h after start, no GNM action	Set GnmStatus = 'KIV'
GCU Reminder	2 days before start	Send reminder email to all GcuAcknowledgement records for this outage
Not Taken Lock	After Not Taken status	Block new outage creation by same Requestor until Not-Taken reason submitted
Project Completion Check	On project close attempt	Validate no open outages remain; warn if found

12. .NET & Entity Framework Core Alignment

12.1 Entity and DbContext Mapping

- Each table maps to a C# entity class in the ICOMS.Domain project. Entities in the config schema are grouped under a ConfigurationDbContext; auth schema tables use IdentityDbContext<ApplicationUser> extended with custom entities.
- Soft-delete is enforced via a global query filter: `modelBuilder.Entity<T>().HasQueryFilter(e => !e.IsDeleted)`.
- Audit interception is implemented via `SaveChangesInterceptor` in EF Core — automatically populates `CreatedAt`, `CreatedBy`, `UpdatedAt`, `UpdatedBy`, and writes to `audit.AuditLog`.
- Enum-like status codes (`OutageTypeCode`, `StatusCode`) are stored as `NVARCHAR` and mapped to C# enum types via value converters for type safety.
- Database migrations are managed via EF Core CLI (`dotnet ef migrations add`). Seed data for roles, permissions, zones, and dropdown values is applied via `IEntityTypeConfiguration<T> HasData()`.

12.2 Blazor & API Layer

- Blazor components consume data via `HttpClient` calling the ASP.NET Core Web API. SignalR hubs notify Blazor clients of real-time status changes (e.g., outage Taken-Active, GCU acknowledgements).
- All API endpoints enforce `[Authorize]` with policy-based role checks aligned to `auth.RolePermissions.FeatureCode`.
- Pagination is handled server-side using LINQ `Skip/Take` with `IQueryable` projections — never client-side loading of full datasets.
- File uploads (SLDs, PDFs, Contingency Plans) are streamed to Azure Blob Storage via a dedicated `FileStorageService`; the URL is stored in the database.

13. Data Governance & Retention

- Soft-delete (IsDeleted = 1) is the default deletion pattern for all user-facing records. Hard deletes are restricted to audit.AuditLog pruning by SysAdmin only.
- Outage records are retained indefinitely; only status changes are tracked via audit.AuditLog.
- Handover records are retained for a minimum of 3 years per TNB operational policy.
- Equipment history (outage list per equipment) shows all outages with status Outage Closed for the last 3 years (all job types except Maintenance) and last 10 years for Maintenance job type only.
- PDPA compliance: Email and PhoneNumber fields are classified as personal data. Access is logged. Data masking is applied in non-production environments.
- Database backups: Full nightly, differential every 6 hours, transaction log every 15 minutes. RPO target: 15 minutes; RTO target: 1 hour.

Appendix A — Complete Table Inventory

#	Table	Schema	Purpose
1	Users	auth	All system users — internal (AD) and external (Membership)
2	Roles	auth	System roles
3	RolePermissions	auth	Feature-level permissions per role
4	UserGcuStations	auth	GCU user to station mapping
5	RoleTransferRequests	auth	User-submitted role/zone change requests
6	Zones	config	Operational zones
7	ZoneLocations	config	State/territory to zone mapping
8	Organisations	config	Internal and external organisations
9	Stations	config	Electrical stations
10	VoltageLevel	config	Voltage levels (132kV, 275kV, 500kV)
11	EquipmentType	config	Equipment types per voltage
12	Equipment	config	Individual equipment items
13	ConflictingLines	config	Line conflict pairs for outage validation
14	DropdownValues	config	Generic configurable dropdown store
15	Projects	config	Outage project records
16	OutageTypeRules	config	Auto-classification rules for outage types
17	AuthorisationPersonnel	config	Pre-approved signatory personnel
18	Outages	dbo	Master outage record
19	OutageAdditionalEquipment	dbo	Additional equipment per outage
20	OutagePic	dbo	Person-in-charge contacts
21	OutageNotifyEmails	dbo	Additional email recipients
22	ChangeRequests	dbo	Outage modification requests
23	OutageOffPoints	dbo	TOMS off-point suggestions per outage
24	GcuAcknowledgements	dbo	GCU user acknowledgement records
25	Authorisations	dbo	GNC Taken-Active/Taken-Completed records
26	SingleLineDiagrams	dbo	Versioned SLD records per station
27	CommissioningMemos	dbo	Commissioning memo per outage
28	HandoverShifts	handover	Shift master record (Morning/Evening/Night)
29	HandoverEntries	handover	Shift detail line items per tab/field
30	AuditLog	audit	Immutable change audit trail
31	SavedReportFilters	audit	User-saved filter presets